

Do IMF Article IV Reports Contain Early-Warning Signals of Financial Crises? A Retrieval-Augmented Approach

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Repository: <https://github.com/Nihad-Isgandarli/imf-article-iv-rag>

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Abstract

The empirical literature on financial-crisis prediction relies almost entirely on numerical indicators. We ask whether the language of IMF Article IV consultation reports carries detectable early-warning content. Using a Retrieval-Augmented Generation (RAG) pipeline, we embed 55 reports from 14 countries, ask each report five standardised concern questions, and have a large language model score the retrieved text on a 0-10 scale per question. The composite 0-50 index separates crisis countries (mean 33.4) from controls (mean 18.8), and predicts a next-year crisis with an AUC of up to 0.87. The signal is concentrated in external vulnerability, which is the only dimension that is overwhelmingly significant ahead of crises ($t = 6.0$, $p < 0.001$). We detail the methodology, the validation of the scores, the results, and the limitations of using a language model as a measurement instrument.

1 Introduction and Problem Definition

When a financial crisis breaks out, observers often note afterwards that the warning signs were "there all along". Most quantitative early-warning systems (EWS) formalise this intuition with macro-financial numbers: debt-to-GDP ratios, reserve adequacy, fiscal deficits, credit growth (Kaminsky et al., 1998; Reinhart and Rogoff, 2009; Frankel and Saravelos, 2012). These models are powerful but they ignore a rich, systematic, and freely available textual record of country risk: the IMF Article IV consultation reports, produced annually for nearly every member country as part of the Fund's surveillance mandate.

Our research question is therefore: do IMF Article IV reports contain detectable early-warning signals of financial crises?

The contribution is to treat the IMF's own prose as data. Where the EWS literature reads the macro-financial numbers, we read the surveillance language: modern retrieval and language models make it feasible to read hundreds of long reports systematically and to convert qualitative concern into a comparable numeric index, rather than relying on ad-hoc keyword counts. To our knowledge this combination (a theory-grounded set of concern questions, retrieval over an embedded corpus, and LLM scoring against an explicit rubric) has not been applied to IMF Article IV reports at scale. Our anchor case is in-sample: in the IMF's reports on Lebanon, concern is already elevated in the mid-2010s and rises steadily, with the 2018 Article IV report scoring 42/50 a year before Lebanon's 2019 banking and financial crisis.

2 Project Plan

Objective. Build an end-to-end, reproducible pipeline that scores IMF Article IV reports for crisis concern and evaluates those scores against realised crises, with a transparent development process on GitHub.

Scope and rubric. The project combines three of the elements required by the course brief, including one advanced component: a large language model (Google Gemini, the advanced component); a non-trivial database (a ChromaDB vector store of embedded report chunks); and advanced data visualisation (an interactive Streamlit dashboard). It is organised as an agentic project, with an `AGENTS.md` file and at least one pull request authored by an AI agent.

Sample design. We deliberately kept the corpus small, just 14 countries and 55 reports (2014-2025), so that each country could be examined closely and so that scoring stayed within the free tier of the language-model API. Nine countries are treatment (crisis) cases and five are controls (Table 1).

Division of labour. Nihad handled data collection (scraping IMF.org), the Reinhart-Rogoff crisis data, and the statistical analysis (hypothesis tests and ROC). Alessandro built the ChromaDB ingestion, the RAG querying module, the scoring system, and the Streamlit dashboard. Vittoria designed and phrased the five questions, validated the scores, and wrote this report.

3 Project Diary

We began by extracting and chunking a single report to fix the text pipeline, then generalised it. Scraping IMF.org proved non-trivial: the site sits behind Akamai bot protection, which blocks plain HTTP requests at both stages. We therefore use Playwright with Chromium to locate the report links on each country page and to download the PDFs through a real browser context, with a manual fallback for Lebanon 2018, which had no standard PDF.

We collected historical crisis dates from the Reinhart-Rogoff dataset (Reinhart and Rogoff, 2009) (coverage to 2016) and added recent crises manually, recording the source of each row in `crises.csv`. Text was embedded locally with `sentence-transformers` and stored in ChromaDB; the RAG query module retrieves the most relevant chunks for a question and sends them to the model with a fixed rubric, requesting a structured JSON answer.

One methodological correction is worth recording. An initial scoring run used placeholder questions for testing; once the finalised five-question specification was ready, the scoring was re-run on all 55 reports and the score columns were renamed to match. The validation, the statistical analysis, and this report all use the re-run scores. We used Codex as an AI agent to open a pull request that updates the project's documentation files; the process is documented in the repository's commit history.

4 Methodology: the Data Science Behind the Project

4.1 Corpus and crisis dating

The corpus is 55 IMF Article IV reports (International Monetary Fund, 2014-2025) across 14 countries (Table 1). Country selection balanced two goals: covering the main types of crisis (currency, banking, sovereign debt) and providing a credible control group. A country is labelled treatment if it experienced a crisis within our window and control otherwise. Controls are actively surveilled economies that were plausibly crisis-free over 2014-2025, so that low scores reflect genuinely calmer reports rather than missing data. Crisis dates combine the Reinhart-Rogoff dataset (Reinhart and Rogoff, 2009) (to 2016) with manual additions for recent episodes (for example Sri Lanka 2022, Ghana 2022, Zambia 2020), each tagged with its source. Control countries are crisis-free within the analysed report window; a few pre-2017 currency flags in the Reinhart-Rogoff data (for example for Poland) fall outside the scored report years and therefore do not enter the statistical tests.

Table 1: Countries in the sample.

Group	Countries	Crisis type(s)
Treatment (9)	Turkey, Argentina, Pakistan, Egypt	currency
	Sri Lanka, Ghana, Zambia, Ecuador	sovereign debt
	Lebanon	banking
Control (5)	Switzerland, Canada, Korea, Poland, Malaysia	none

4.2 The five concern questions

Each question maps to one of the standard categories in the EWS literature (fiscal, financial-sector, external, momentum, and policy stance) and plays two roles in the pipeline: a retrieval query (embedded and matched against report chunks) and a scoring question (the instruction the model answers on the retrieved text). The full specification is in `docs/rag_questions.md`; the scoring questions are reproduced verbatim below.

Q1 (fiscal)

To what extent does the report express concern about fiscal sustainability: rising public debt, large or widening fiscal deficits, or growing gross financing needs?

Q2 (banking)

To what extent does the report flag vulnerabilities in the banking or financial sector: rising non-performing loans, thin capital buffers, rapid credit growth, or funding/liquidity stress?

Q3 (policy urgency)

How urgent is the policy stance: does the report call for immediate, urgent, or front-loaded action, or warn that delay carries significant risks?

Q4 (deterioration)

Does the report describe a deterioration relative to previous years in key macro-financial indicators (growth, reserves, fiscal or external balances, asset quality)?

Q5 (external)

To what extent does the report flag external vulnerabilities: capital outflows or sudden

stops, exchange-rate/currency pressure, low or falling reserves, or current-account / balance-of-payments stress?

4.3 Pipeline

The pipeline is a standard RAG architecture (Lewis et al., 2020):

PDF $\rightarrow$ text $\rightarrow$ chunks $\rightarrow$ embeddings $\rightarrow$ ChromaDB $\rightarrow$ retrieve top- k $\rightarrow$ LLM score.

Reports are split into 500-word chunks with a 50-word overlap to preserve context across chunk boundaries. Chunks are embedded with the local all-MiniLM-L6-v2 model (Reimers and Gurevych, 2019), which produces 384-dimensional embeddings locally, with no API call. Embeddings are stored in a persistent ChromaDB collection; for each question the system retrieves the five nearest chunks (ChromaDB’s default L2 metric). Generation uses Google Gemini (gemini-3.1-flash-lite) at a low temperature of 0.2; the model sees only the retrieved chunks and the question, and returns structured JSON with an answer, an integer concern score, and a short reasoning. Confining the model to the retrieved passages keeps each score anchored in the report itself rather than in the model’s background knowledge, which guards against hallucination, while any reply that is not valid JSON is caught and handled rather than allowed to break the run.

4.4 Scoring rubric and aggregation

Every question uses the same 0-10 rubric (Table 2). For report i and question j , let $s_{ij} \in \{0, \dots, 10\}$. The default, equal-weight Concern Index is the equal-weight sum $C_i = \sum_{j=1}^5 s_{ij} \in [0, 50]$ (equivalently $C_i/5$ for a 0-10 average); a weighted variant is a natural robustness check given the differing discriminating power of the dimensions (Section 5).

Table 2: Scoring rubric, applied identically to every question.

Score	Meaning
0-2	Not raised, or explicitly reassuring / under control
3-4	Mild, routine concern; risks noted but described as manageable
5-6	Moderate concern; clear risks, monitoring or gradual action advised
7-8	High concern; prominent risks, strong language, significant vulnerabilities
9-10	Severe/acute concern; crisis-level language, imminent threat, urgent action demanded

5 Results

5.1 Composite separation

The composite 0-50 index separates crisis countries from controls (Table 3): treatment countries average 33.4 versus 18.8 for controls, and no control report exceeds 24 (Malaysia 2023) while crisis-window reports reach 47 (Lebanon 2023). The highest readings are Lebanon 2023 (47), Sri Lanka 2021 (44, one year before the 2022 default), and Lebanon 2018 and 2019 (42).

Table 3: Concern Index (0-50) by country group, all report-years.

Group	N	Mean	SD	Min	Max
Treatment (crisis)	30	33.4	7.1	21	47
Control	25	18.8	2.3	14	24

5.2 Early-warning test: pre-crisis versus calm years

We compare reports from the one-to-two years before a crisis ($n = 9$) against all calm reports ($n = 46$), per dimension, with Welch’s t -test (Table 4). Only two dimensions differ significantly at the 5% level: external vulnerability (q5), which is by far the strongest (mean 7.67 versus 4.46; $t = 6.0$, $p < 0.001$), and deterioration (q4; 6.67 versus 5.24; $t = 2.37$, $p = 0.034$). Fiscal, banking, and policy urgency do not reach significance on this test. External vulnerability is thus the clearest pre-crisis warning in the IMF’s language.

Table 4: Pre-crisis versus calm-year scores, by dimension.

Dimension	Mean pre	Mean calm	Diff	t	p
q1 fiscal	6.67	5.35	1.32	1.51	0.159
q2 banking	5.56	4.37	1.19	1.19	0.264
q3 policy	7.00	6.00	1.00	1.77	0.101
q4 deterioration	6.67	5.24	1.43	2.37	0.034
q5 external	7.67	4.46	3.21	6.00	<0.001

5.3 Predictive power: ROC

Treating each dimension’s score as a classifier for whether a crisis occurs the following year (Figure 1), all five beat random (AUC 0.50). External vulnerability is again the best predictor (AUC 0.87), followed by banking (0.78), policy urgency (0.76), fiscal (0.72), and deterioration (0.70). Banking’s relatively high AUC is notable because its group means barely differ (Section 7); with only five to six crisis events in the panel, these AUC estimates are necessarily noisy. A complementary cut by crisis status shows the same ordering on every dimension: crisis-year reports score highest, controls lowest, and treatment countries’ calm years in between, consistent with concern rising as a country approaches a crisis.

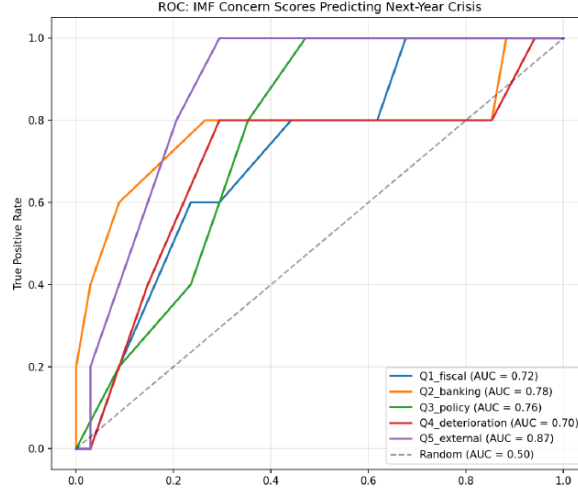


Figure 1: ROC curves: each concern dimension as a classifier for a next-year crisis.

5.4 Anchor case: Lebanon

Lebanon illustrates the trajectory (Figure 2): concern is already elevated in the mid-2010s, rises through the decade, and is very high by the 2019 banking crisis, with external vulnerability leading, mirroring the panel-wide finding.

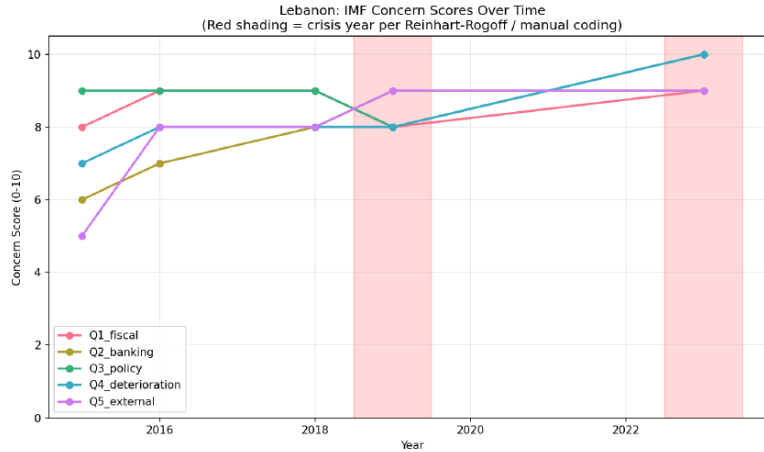


Figure 2: Lebanon: concern scores over time; red shading marks the crisis year.

6 Validation

All 275 model responses (55 reports times 5 questions) were screened for internal consistency between the assigned 0-10 score, the model's stated reasoning, and the rubric in `docs/rag_questions.md`. Of these, 274 were retained, 1 was flagged for reconsideration (Ecuador 2024, q1: the IMF classifies

the debt risk as High, yet the score is a moderate 6), and 0 were revised. The calibration is consistent across every score band, from reassuring language at the low end to crisis-level language at the high end. In addition, a random sample of fifteen answers was read against the source reports to check that the model’s reasoning reflected the underlying text; in this sample the assigned scores were consistent with the reports. This verifies internal consistency and adds a spot check against the source text; it is not a full independent re-scoring of every report.

7 Reflection on the Economic Results

Three observations stand out. First, the IMF’s prose does carry early-warning content, and the signal is concentrated in external vulnerability: q5 separates crisis from calm reports most sharply, is the only overwhelmingly significant pre-crisis dimension, and is the strongest next-year predictor (AUC 0.87). The economic reading is intuitive given our sample, where currency and balance-of-payments stress is the most common trigger: the Fund’s language flags external fragility most clearly before crises, with deterioration (momentum) as a secondary warning. This echoes the macro-financial EWS literature, where reserves and the real exchange rate, both external indicators, are repeatedly found to be the most useful leading indicators (Frankel and Saravelos, 2012); our contribution is to recover the same signal from the IMF’s words rather than its numbers.

Second, the dimensions disagree across tests in an informative way. Policy urgency (q3) and banking (q2) are weak on both the pre-crisis t-test and on mean differences, yet banking ranks second on AUC, a reminder that a shift in means and a rank-based classifier need not agree when there are only five to six crisis events. Policy urgency is elevated even for controls because it measures the forcefulness of IMF language, which rises whenever the Fund urges structural reform, crisis or not. The textbook hard-versus-soft split therefore does not map cleanly onto signal strength: external (hard) is strongest, deterioration (soft) is the clear secondary, and banking (hard) is weak in our crisis mix, since only Lebanon is primarily a banking crisis.

Third, the apparent false positives among controls are economically real rather than noise (Switzerland 2023 reflects the UBS-Credit Suisse episode, Malaysia a post-pandemic debt build-up).

7.1 Limitations and threats to validity

Several caveats temper these results.

- (i) Few events and low power. The panel contains only about five to six crisis-years and nine pre-crisis observations, so inferential estimates, especially the AUCs, are noisy and sensitive to individual cases such as Lebanon.
- (ii) Endogeneity and contemporaneity. The IMF may voice concern only once distress is visible, so part of the prediction is contemporaneous; the pre-crisis test, which uses only the one-to-two years before a crisis, mitigates but does not eliminate this.
- (iii) The language model as a measurement instrument. Scores depend on the model version, the prompt, and the temperature, and are not independently calibrated against expert human scoring.
- (iv) Retrieval coverage. Each score reflects only the chunks retrieved for that question.
- (v) Crisis dating. The manual crisis additions involve judgement on exact years.

- (vi) Language and source. The analysis is English-only and specific to the IMF’s institutional style.

8 Lessons Learned

- Specification before scale. Re-running the scoring after the questions were finalised cost a full pass; agreeing the exact questions first would have avoided it. We added a documentation rule to keep the questions in sync across the code, the README, and the specification.
- Reproducibility is a deliverable. A clean `requirements.txt` and an accurate `AGENTS.md` matter as much as the code; we fixed both so the project installs and runs from scratch.
- The soft signal is the noisy one. Tone-based dimensions such as policy urgency discriminate worse than the strongest macro dimension, external vulnerability, which is an argument for weighting in any text-based EWS.
- A language model is a measurement instrument, not an oracle. It must be constrained (retrieved context, fixed rubric, low temperature) and then validated, first for internal consistency at scale and then against the source text on a human-checked sample.

9 Use of Generative AI

Generative-AI tools were used during the project and are acknowledged here. Code was written with the assistance of Claude, and each author can explain their own contribution. Claude was also used for design discussion, for the consistency screening behind Section 6, and for drafting and structuring this report. The required AI-agent pull request was created with Codex, which produced a documentation-synchronisation branch following the conventions in `AGENTS.md`; the pull request was reviewed and merged by a human author.

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